Fractional integration associated with Zygmund dilations

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Abstract

First, we study a family of fractional integral operator defined as

$$\mathbf{I}_{\alpha\beta\vartheta} f(u,v,t) \;=\; \iiint_{\mathbb{R}^{2n+1}} f(\xi,\eta,\tau)\mathbf{V}^{\alpha\beta\vartheta}[(u,v,t)\odot(\xi,\eta,\tau)^{-1}]d\xi d\eta d\tau$$

where $\odot$ denotes the multiplication law on a real Heisenberg group.

$\mathbf{V}^{\alpha\beta\vartheta}$ is a distribution in $\mathbb{R}^{2n+1}$ satisfying Zygmund dilations. A characterization is established between $\mathbf{I}_{\alpha\beta\vartheta}\colon \mathbf{L}^p(\mathbb{R}^{2n+1}) \longrightarrow \mathbf{L}^q(\mathbb{R}^{2n+1})$ and necessary constraints consisting of $\alpha,\beta\in\mathbb{R}$ and $\vartheta\geq 0$ for $1<p<q<\infty$.

Moreover, denote $\mathbf{R}=\mathbf{Q}\times\mathbf{P}\times I\subset\mathbb{R}^n\times\mathbb{R}^n\times\mathbb{R}$: I is an interval and $\mathbf{Q},\mathbf{P}$ are cubes parallel to the coordinates. For $\alpha,\beta\in\mathbb{R}$, we consider

$$\mathbf{M}_{\alpha\beta} f(u,v,t) \;=\; \sup_{\mathbf{R}\ni(0,0,0):\mathbf{vol}\{I\}=\mathbf{vol}\{\mathbf{Q}\}^{\frac{1}{n}}\mathbf{vol}\{\mathbf{P}\}^{\frac{1}{n}}} \mathbf{vol}\{\mathbf{Q}\}^{\frac{\alpha}{n}-1}\mathbf{vol}\{\mathbf{P}\}^{\frac{\alpha}{n}-1}\mathbf{vol}\{I\}^{\beta-1}\iiint_{\mathbf{R}}\Big|f[(u,v,t)\odot(\xi,\eta,\tau)^{-1}]\Big|d\xi d\eta d\tau.$$

We show $\mathbf{M}_{\alpha\beta}\colon \mathbf{L}^p(\mathbb{R}^{2n+1}) \longrightarrow \mathbf{L}^q(\mathbb{R}^{2n+1})$ for $1<p\leq q<\infty$ whenever $\frac{\alpha+\beta}{n+1}=\frac{1}{p}-\frac{1}{q}$.

1 Introduction

A fractional integral operator $\mathbf{T_a}$ is initially defined on $\mathbb{R}^{\mathbf{N}}$ as

$$\mathbf{T_a} f(x) \;=\; \int_{\mathbb{R}^{\mathbf{N}}} f(y)\left[\frac{1}{|x-y|}\right]^{\mathbf{N}-\mathbf{a}} dy, \qquad 0<\mathbf{a}<\mathbf{N}. \tag{1. 1}$$

In 1928, Hardy and Littlewood [1] have obtained an regularity theorem for $\mathbf{T_a}$ when $\mathbf{N}=1$. Ten years later, Sobolev [2] made extensions on every higher dimensional space.

$\diamond$ Throughout, $\mathfrak{B}>0$ is regarded as a generic constant depending on its sub-indices.

Hardy-Littlewood-Sobolev theorem *Let $\mathbf{T_a}$ defined in (1. 1) for $0<\mathbf{a}<\mathbf{N}$. We have*

$$\begin{array}{c}\Big\|\mathbf{T_a} f\Big\|_{\mathbf{L}^q(\mathbb{R}^{\mathbf{N}})} \;\leq\; \mathfrak{B}_{p\,q}\,\Big\|f\Big\|_{\mathbf{L}^p(\mathbb{R}^{\mathbf{N}})}, \qquad 1<p<q<\infty \\ \textit{if and only if} \qquad \dfrac{\mathbf{a}}{\mathbf{N}} \;=\; \dfrac{1}{p}-\dfrac{1}{q}.\end{array} \tag{1. 2}$$

This classical result was first re-investigated by Folland and Stein [4] on Heisenberg groups.

In this paper, $\odot$ denotes a multiplication law:

$$(u,v,t)\odot(\xi,\eta,\tau) ~=~ \Big[u+\xi, v+\eta, t+\tau+\mu(u\cdot\eta-v\cdot\xi)\Big], \qquad \mu\in\mathbb{R} \tag{1. 3}$$

for every $(u,v,t)\in\mathbb{R}^n\times\mathbb{R}^n\times\mathbb{R}$ and $(\xi,\eta,\tau)^{-1}=(-\xi,-\eta,-\tau)\in\mathbb{R}^n\times\mathbb{R}^n\times\mathbb{R}$.

Let $0<\delta<n+1$. Consider

$$\mathbf{S}_\delta f(u,v,t) ~=~ \iiint_{\mathbb{R}^{2n+1}} f(\xi,\eta,\tau)\Omega^\delta\Big[(u,v,t)\odot(\xi,\eta,\tau)^{-1}\Big]d\xi d\eta d\tau. \tag{1. 4}$$

Ω^δ is a distribution in $\mathbb{R}^{2n+1}$ agree with

$$\Omega^\delta(u,v,t) ~=~ \left[\frac{1}{|u|^2+|v|^2+|t|}\right]^{n+1-\delta}, \qquad (u,v,t)\neq(0,0,0). \tag{1. 5}$$

Folland-Stein theorem *Let* $\mathbf{S}_\delta$ *defined in (1. 4)-(1. 5) for* $0<\delta<n+1$. *We have*

$$\begin{gathered}\Big\|\mathbf{S}_\delta f\Big\|_{\mathbf{L}^q(\mathbb{R}^{n+1})} ~\leq~ \mathfrak{B}_{p\,q}~\Big\|f\Big\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})}, \qquad 1<p<q<\infty \\ \textit{if and only if} \qquad \frac{\delta}{n+1} ~=~ \frac{1}{p}-\frac{1}{q}.\end{gathered} \tag{1. 6}$$

The best constant for the $\mathbf{L}^p\longrightarrow\mathbf{L}^q$-norm inequality in (1. 6) is found by Frank and Lieb [13]. A discrete analogue of this result has been obtained by Pierce [14]. Recently, the regarding commutator estimates are established by Fanelli and Roncal [15].

Next, we give a multi-parameter extension to **Folland-Stein theorem** by replacing Ω^δ with a larger kernel having singularity on every coordinate subspace. From (1. 5), we have

$$\Omega^\delta(u,v,t) ~\leq~ \left[\frac{1}{|u||v|+|t|}\right]^{n+1-\delta}, \qquad (u,t)\neq(0,0) \quad\text{or}\quad (v,t)\neq(0,0).$$

A direct computation shows

$$\begin{aligned}\left[\frac{1}{|u||v|+|t|}\right]^{n+1-\delta} ~&\approx~ \left[\frac{1}{|u|^2|v|^2+t^2}\right]^{\frac{n+1}{2}-\frac{\delta}{2}} \\ &=~ |u|^{\frac{\delta}{2}-\frac{n+1}{2}}|v|^{\frac{\delta}{2}-\frac{n+1}{2}}|t|^{\frac{\delta}{2}-\frac{n+1}{2}}\left[\frac{|u||v||t|}{|u|^2|v|^2+t^2}\right]^{\frac{n+1}{2}-\frac{\delta}{2}} \\ &=~ |u|^{\left[\frac{\delta}{2}+\frac{n-1}{2}\right]-n}|v|^{\left[\frac{\delta}{2}+\frac{n-1}{2}\right]-n}|t|^{\left[\frac{\delta}{2}-\frac{n-1}{2}\right]-1}\left[\frac{|u||v|}{|t|}+\frac{|t|}{|u||v|}\right]^{-\left[\frac{n+1}{2}-\frac{\delta}{2}\right]}\end{aligned}$$

whenever $(u,t)\neq(0,0)$ or $(v,t)\neq(0,0)$. Above estimates lead us to the following assertion. Let $\alpha,\beta\in\mathbb{R}$ and $\vartheta\geq0$. $\mathbf{V}^{\alpha\beta\vartheta}$ is a distribution in $\mathbb{R}^{2n+1}$ agree with

$$\mathbf{V}^{\alpha\beta\vartheta}(u,v,t) ~=~ |u|^{\alpha-n}|v|^{\alpha-n}|t|^{\beta-1}\left[\frac{|u||v|}{|t|}+\frac{|t|}{|u||v|}\right]^{-\vartheta}, \qquad u\neq0, v\neq0, t\neq0. \tag{1. 7}$$

Remark 1.1. *By taking into account* $\alpha = \frac{\delta}{2} + \frac{n-1}{2}$, $\beta = \frac{\delta}{2} - \frac{n-1}{2}$ *and* $\vartheta = \frac{n+1}{2} - \frac{\delta}{2}$ *for* $0 < \delta < n+1$, *we find* $\alpha > n\beta$ *and* $\vartheta = \frac{n+1}{2} - \frac{\delta}{2} > \frac{\alpha - n\beta}{n+1}$.

Define

$$\mathbf{I}_{\alpha\beta\vartheta} f(u,v,t) \; = \; \iiint_{\mathbb{R}^{2n+1}} f(\xi,\eta,\tau)\, \mathbf{V}^{\alpha\beta\vartheta}\Big[(u,v,t)\odot(\xi,\eta,\tau)^{-1}\Big] d\xi d\eta d\tau. \qquad (1.\ 8)$$

Observe that

$$\mathbf{V}^{\alpha\beta\vartheta}\Big[(ru,sv,rst)\odot(r\xi,s\eta,rs\tau)^{-1}\Big] \; = \; r^{\alpha+\beta-n-1}s^{\alpha+\beta-n-1}\mathbf{V}^{\alpha\beta\vartheta}\Big[(u,v,t)\odot(\xi,\eta,\tau)^{-1}\Big], \quad r,s>0. \quad (1.\ 9)$$

A convolution operator of this type is said to be associated with Zygmund dilations. Certain singular integral operators defined on a Heisenberg group associated with Zygmund dilations have been systematically studied, for example by Phong and Stein [5], Ricci and Stein [6] and Müller, Ricci and Stein [7]. Much less is known for fractional integrals in this direction.

Theorem One *Let* $\mathbf{I}_{\alpha\beta\vartheta}$ *defined in (1. 7)-(1. 8) for* $\alpha, \beta \in \mathbb{R}$ *and* $\vartheta \geq 0$. *We have*

$$\begin{array}{c} \Big\|\mathbf{I}_{\alpha\beta\vartheta} f\Big\|_{\mathbf{L}^q(\mathbb{R}^{2n+1})} \; \leq \; \mathfrak{B}_{p\,q}\; \Big\|f\Big\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})}, \qquad 1 < p < q < \infty \\[2ex] \textit{if and only if} \qquad \vartheta \geq \dfrac{|\alpha - n\beta|}{n+1}, \qquad \dfrac{\alpha+\beta}{n+1} \; = \; \dfrac{1}{p} - \dfrac{1}{q}. \end{array} \qquad (1.\ 10)$$

Denote $\mathbf{R} = \mathbf{Q}\times\mathbf{P}\times I \subset \mathbb{R}^n\times\mathbb{R}^n\times\mathbb{R}$: I is an interval and $\mathbf{Q}, \mathbf{P}$ are cubes parallel to the coordinates. For $\alpha, \beta \in \mathbb{R}$, we define

$$\begin{array}{l} \mathbf{M}_{\alpha\beta} f(u,v,t) \; = \\ \displaystyle\sup_{\mathbf{R}\ni(0,0,0):\mathbf{vol}\{I\}=\mathbf{vol}\{\mathbf{Q}\}^{\frac{1}{n}}\mathbf{vol}\{\mathbf{P}\}^{\frac{1}{n}}} \mathbf{vol}\{\mathbf{Q}\}^{\frac{\alpha}{n}-1}\mathbf{vol}\{\mathbf{P}\}^{\frac{\alpha}{n}-1}\mathbf{vol}\{I\}^{\beta-1} \iiint_{\mathbf{R}} \Big|f[(u,v,t)\odot(\xi,\eta,\tau)^{-1}]\Big| d\xi d\eta d\tau. \end{array} \qquad (1.\ 11)$$

This is known as the fractional maximal function associated with Zygmund dilations defined on a Heisenberg group. For $\mathbf{M}_{\alpha\beta}$ defined in the Euclidean space, in particular at $\alpha = \beta = 0$, the regarding $\mathbf{L}^p$- theorem and its weighted analogue have been well established. See the papers by Ricci and Stein [6] and Fefferman and Pipher [10].

Theorem Two *Let* $\mathbf{M}_{\alpha\beta}$ *defined in (1. 11) for* $\alpha, \beta \in \mathbb{R}$. *We have*

$$\begin{array}{c} \Big\|\mathbf{M}_{\alpha\beta} f\Big\|_{\mathbf{L}^q(\mathbb{R}^{2n+1})} \; \leq \; \mathfrak{B}_{p\,q}\; \Big\|f\Big\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})}, \qquad 1 < p \leq q < \infty \\[2ex] \textit{whenever} \qquad \dfrac{\alpha+\beta}{n+1} \; = \; \dfrac{1}{p} - \dfrac{1}{q}. \end{array} \qquad (1.\ 12)$$

Later, we shall find

$$\mathbf{M}_{\alpha\beta} f(u,v,t) \; \leq \; \mathbf{M}_\gamma f(u,v,t), \qquad \gamma = \tfrac{\alpha+\beta}{n+1} \qquad (1.\ 13)$$

for which $\mathbf{M}_\gamma, 0 \leq \gamma < 1$ is defined as

$$\mathbf{M}_\gamma f(u,v,t) \; = \; \sup_{\mathbf{R}\ni(0,0,0)} \mathbf{vol}\{\mathbf{R}\}^{\gamma-1} \iiint_{\mathbf{R}} \Big|f[(u,v,t)\odot(\xi,\eta,\tau)^{-1}]\Big| d\xi d\eta d\tau. \qquad (1.\ 14)$$

Remark 1.2. *The supremum in (1. 14) is taking over any rectangle* $\mathbf{R} \subset \mathbb{R}^{2n+1}$ *parallel to the coordinates.*

$\mathbf{M}_0$ in (1. 14) is known as the strong maximal operator defined on Heisenberg groups. The $\mathbf{L}^p$-boundedness of $\mathbf{M}_0$ has been proved by M. Christ [11] using a mixture of techniques developed previously by Ricci and Stein [8]-[9] and Christ [12] for singular integrals defined on sub-manifolds within a general setting of Nilpotent Lie groups. **Theorem Two** will be a corollary of the next result.

Theorem Three *Let* $\mathbf{M}_\gamma$ *defined in (1. 14) for* $0 \leq \gamma < 1$. *We have*

$$\begin{array}{c} \left\|\mathbf{M}_\gamma f\right\|_{\mathbf{L}^q(\mathbb{R}^{2n+1})} \leq \mathfrak{B}_{p\,q} \left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})}, \qquad 1 < p \leq q < \infty \\ \textit{whenever} \qquad \gamma = \frac{1}{p} - \frac{1}{q}. \end{array} \tag{1. 15}$$

The remaining paper is organized as follows. In section 2, we show that two constraints consisting of α, β, ϑ, p, q in (1. 10) are necessary conditions. In section 3, we prove their sufficiency and finish the proof of **Theorem One**. In section 4, we prove **Theorem Three** which implies **Theorem Two** with a new approach by applying a geometric covering lemma due to Córdoba and Fefferman [3].

2 Some necessary constraints

Let $\mathbf{I}_{\alpha\beta\vartheta}$ defined in (1. 7)-(1. 8). By changing variable $\tau \longrightarrow \tau - \mu(u\cdot\eta - v\cdot\xi)$, we find

$$\begin{aligned} \mathbf{I}_{\alpha\beta\vartheta} f(u,v,t) &= \iiint_{\mathbb{R}^{2n+1}} f(\xi,\eta,\tau)\, \mathbf{V}^{\alpha\beta\vartheta}\Big[u-\xi, v-\eta, t-\tau-\mu(u\cdot\eta - v\cdot\xi)\Big] d\xi d\eta d\tau \\ &= \iiint_{\mathbb{R}^{2n+1}} f(\xi,\eta,\tau-\mu(u\cdot\eta - v\cdot\xi))\, \mathbf{V}^{\alpha\beta\vartheta}(u-\xi, v-\eta, t-\tau) d\xi d\eta d\tau \\ &= \iiint_{\mathbb{R}^{2n+1}} f(\xi,\eta,\tau-\mu(u\cdot\eta - v\cdot\xi)) \\ &\qquad |u-\xi|^{\alpha-n}|v-\eta|^{\alpha-n}|t-\tau|^{\beta-1}\left[\frac{|u-\xi||v-\eta|}{|t-\tau|} + \frac{|t-\tau|}{|u-\xi||v-\eta|}\right]^{-\vartheta} d\xi d\eta d\tau. \end{aligned} \tag{2. 1}$$

Consider a more general situation by replacing $\mathbf{V}^{\alpha\beta\vartheta}(u,v,t)$ with

$$|u|^{\alpha_1-n}|v|^{\alpha_2-n}|t|^{\beta-1}\left[\frac{|u||v|}{|t|} + \frac{|t|}{|u||v|}\right]^{-\vartheta}, \qquad \alpha_1,\alpha_2,\beta \in \mathbb{R}, \qquad \vartheta \geq 0 \tag{2. 2}$$

for $u \neq 0, v \neq 0, t \neq 0$.

Let $(u,v,t) \longrightarrow (ru, sv, rs\lambda t)$ and $(\xi,\eta,\tau) \longrightarrow (r\xi, s\eta, rs\lambda\tau)$ for $r,s > 0$ and $0 < \lambda < 1$ or $\lambda > 1$.

We have

$$\left\{\iiint_{\mathbb{R}^{2n+1}}\left\{\iiint_{\mathbb{R}^{2n+1}} f\left[r^{-1}\xi, s^{-1}\eta, r^{-1}s^{-1}\lambda^{-1}[\tau-\mu\lambda(u\cdot\eta-v\cdot\xi)]\right]\right.\right.$$
$$\left.\left.|u-\xi|^{\alpha_1-n}|v-\eta|^{\alpha_2-n}|t-\tau|^{\beta-1}\left[\frac{|u-\xi||v-\eta|}{|t-\tau|}+\frac{|t-\tau|}{|u-\xi||v-\eta|}\right]^{-\vartheta}d\xi d\eta d\tau\right\}^q dudvdt\right\}^{\frac{1}{q}}$$

$$= r^{\alpha_1+\beta}s^{\alpha_2+\beta}r^{\frac{n+1}{q}}s^{\frac{n+1}{q}}\lambda^{\beta}\lambda^{\frac{1}{q}}\left\{\iiint_{\mathbb{R}^{2n+1}}\left\{\iiint_{\mathbb{R}^{2n+1}} f(\xi,\eta,\tau-\mu(u\cdot\eta-v\cdot\xi))\right.\right.$$
$$\left.\left.|u-\xi|^{\alpha_1-n}|v-\eta|^{\alpha_2-n}|t-\tau|^{\beta-1}\left[\frac{|u-\xi||v-\eta|}{\lambda|t-\tau|}+\frac{\lambda|t-\tau|}{|u-\xi||v-\eta|}\right]^{-\vartheta}d\xi d\eta d\tau\right\}^q dudvdt\right\}^{\frac{1}{q}} \qquad (2.\ 3)$$

$$\geq r^{\alpha_1+\beta}s^{\alpha_2+\beta}r^{\frac{n+1}{q}}s^{\frac{n+1}{q}}\lambda^{\beta}\lambda^{\frac{1}{q}}\begin{cases}\lambda^{\vartheta}, & 0<\lambda<1,\\ \lambda^{-\vartheta}, & \lambda>1\end{cases}$$
$$\left\{\iiint_{\mathbb{R}^{2n+1}}\left\{\iiint_{\mathbb{R}^{2n+1}} f(\xi,\eta,\tau-\mu(u\cdot\eta-v\cdot\xi))\right.\right.$$
$$\left.\left.|u-\xi|^{\alpha_1-n}|v-\eta|^{\alpha_2-n}|t-\tau|^{\beta-1}\left[\frac{|u-\xi||v-\eta|}{|t-\tau|}+\frac{|t-\tau|}{|u-\xi||v-\eta|}\right]^{-\vartheta}d\xi d\eta d\tau\right\}^q dudvdt\right\}^{\frac{1}{q}}.$$

The $\mathbf{L}^p \longrightarrow \mathbf{L}^q$-norm inequality in (1. 15) implies that the last line of (2. 3) is bounded by

$$\left\{\iiint_{\mathbb{R}^{2n+1}}\left|f\left(r^{-1}\xi, s^{-1}\eta, r^{-1}s^{-1}\lambda^{-1}\tau\right)\right|^p d\xi d\eta d\tau\right\}^{\frac{1}{p}} = r^{\frac{n+1}{p}}s^{\frac{n+1}{p}}\lambda^{\frac{1}{p}}\left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})}. \qquad (2.\ 4)$$

This must be true for every $r,s>0$ and $0<\lambda<1$ or $\lambda>1$. We necessarily have

$$\frac{\alpha_1+\beta}{n+1} = \frac{1}{p}-\frac{1}{q} = \frac{\alpha_2+\beta}{n+1}, \qquad \beta+\vartheta \geq \frac{1}{p}-\frac{1}{q} \qquad \text{or} \qquad \beta-\vartheta \leq \frac{1}{p}-\frac{1}{q}. \qquad (2.\ 5)$$

The first constraint in (2. 5) forces us to have $\alpha_1=\alpha_2$. Therefore, write

$$\frac{\alpha+\beta}{n+1} = \frac{1}{p}-\frac{1}{q}, \qquad \alpha=\alpha_1=\alpha_2. \qquad (2.\ 6)$$

By bringing (2. 6) to the two inequalities in (2. 5), we find

$$\vartheta \geq \beta-\frac{\alpha+\beta}{n+1} = \frac{n\beta-\alpha}{n+1} \qquad \text{or} \qquad \vartheta \geq \frac{\alpha+\beta}{n+1}-\beta = \frac{\alpha-n\beta}{n+1}. \qquad (2.\ 7)$$

Together, we conclude $\vartheta \geq \frac{|\alpha-n\beta|}{n+1}$.

3 Proof of Theorem One

Given $\alpha,\beta\in\mathbb{R}$ and $\vartheta\geq\frac{|\alpha-n\beta|}{n+1}$, $\mathbf{V}^{\alpha\beta\vartheta}$ is a distribution in $\mathbb{R}^{2n+1}$ agree with $\mathbf{V}^{\alpha\beta\vartheta}(u,v,t)$ in (1. 7) whenever $u\neq0, v\neq0, t\neq0$.

Suppose $\alpha \geq n\beta$. We have $\frac{|\alpha-n\beta|}{n+1} = \frac{\alpha-n\beta}{n+1}$ and

$$\begin{aligned}
\mathbf{V}^{\alpha\beta\vartheta}(\xi,\eta,\tau) \ &\leq \ |u|^{\alpha-n}|v|^{\alpha-n}|t|^{\beta-1}\left[\frac{|u||v|}{|t|}+\frac{|t|}{|u||v|}\right]^{-\frac{\alpha-n\beta}{n+1}} \\
&\leq \ |u|^{\alpha-n}|v|^{\alpha-n}|t|^{\beta-1}\left[\frac{|u||v|}{|t|}\right]^{-\frac{\alpha-n\beta}{n+1}} \qquad (3.\ 1)\\
&= \ |u|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}|v|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}|t|^{\frac{\alpha+\beta}{n+1}-1}, \qquad u \neq 0, v \neq 0, t \neq 0.
\end{aligned}$$

Suppose $\alpha \leq n\beta$. We find $\frac{|\alpha-n\beta|}{n+1} = \frac{n\beta-\alpha}{n+1}$ and

$$\begin{aligned}
\mathbf{V}^{\alpha\beta\vartheta}(u,v,t) \ &\leq \ |u|^{\alpha-n}|v|^{\alpha-n}|t|^{\beta-1}\left[\frac{|u||v|}{|t|}+\frac{|t|}{|u||v|}\right]^{\frac{\alpha-n\beta}{n+1}} \\
&\leq \ |u|^{\alpha-n}|v|^{\alpha-n}|t|^{\beta-1}\left[\frac{|t|}{|u||v|}\right]^{\frac{\alpha-n\beta}{n+1}} \qquad (3.\ 2)\\
&= \ |u|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}|v|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}|t|^{\frac{\alpha+\beta}{n+1}-1}, \qquad u \neq 0, v \neq 0, t \neq 0.
\end{aligned}$$

Let $\mathbf{I}_{\alpha\beta\vartheta} f$ defined in (1. 7)-(1. 8) and

$$\frac{\alpha+\beta}{n+1} \ = \ \frac{1}{p}-\frac{1}{q}, \qquad 1<p<q<\infty. \qquad (3.\ 3)$$

By changing variable $\tau \longrightarrow \tau+\mu(u\cdot\eta-v\cdot\xi)$, we have

$$\begin{aligned}
\mathbf{I}_{\alpha\beta\vartheta} f(u,v,t) \ &= \ \iiint_{\mathbb{R}^{2n+1}} f\left(\xi,\eta,\tau-\mu(u\cdot\eta-v\cdot\xi)\right) \\
&\qquad |u-\xi|^{\alpha-n}|v-\eta|^{\alpha-n}|t-\tau|^{\beta-1}\left[\frac{|u-\xi||v-\eta|}{|t-\tau|}+\frac{|t-\tau|}{|u-\xi||v-\eta|}\right]^{-\vartheta} d\xi d\eta d\tau \qquad (3.\ 4)\\
&\leq \ \iiint_{\mathbb{R}^{2n+1}} f\left(\xi,\eta,\tau-\mu(u\cdot\eta-v\cdot\xi)\right) \\
&\qquad |u-\xi|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}|v-\eta|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}|t-\tau|^{\frac{\alpha+\beta}{n+1}-1} d\xi d\eta d\tau \qquad \text{by (3. 1)-(3. 2).}
\end{aligned}$$

Because $\mathbf{V}^{\alpha\beta\vartheta}$ is positive definite, it is suffice to assert $f \geq 0$. Define

$$\mathbf{F}_{\alpha\beta}(\xi,\eta,u,v,t) \ = \ \int_{\mathbb{R}} f\left(\xi,\eta,\tau-\mu\left(u\cdot\eta-v\cdot\xi\right)\right)|t-\tau|^{\frac{\alpha+\beta}{n+1}-1}d\tau. \qquad (3.\ 5)$$

From (3. 4)-(3. 5), we find

$$\mathbf{I}_{\alpha\beta\vartheta} f(u,v,t) \ \leq \ \iint_{\mathbb{R}^{2n}} |u-\xi|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}|v-\eta|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}\mathbf{F}_{\alpha\beta}(\xi,\eta,u,v,t)d\xi d\eta. \qquad (3.\ 6)$$

Recall the **Hardy-Littlewood-Sobolev theorem** stated in the beginning of this paper. By applying (1. 2) with $\mathbf{a} = \frac{\alpha+\beta}{n+1}$ and $\mathbf{N} = 1$, we have

$$\begin{aligned}\left\{\int_{\mathbb{R}} \mathbf{F}^q_{\alpha\beta}(\xi,\eta,u,v,t)dt\right\}^{\frac{1}{q}} &\le \mathfrak{B}_{p\,q}\left\{\int_{\mathbb{R}}\Big[f\left(\xi,\eta,t-\mu\left(u\cdot\eta-v\cdot\xi\right)\right)\Big]^p dt\right\}^{\frac{1}{p}} \\ &= \mathfrak{B}_{p\,q}\left\|f(\xi,\eta,\cdot)\right\|_{\mathbf{L}^p(\mathbb{R})}\end{aligned} \qquad (3.\ 7)$$

regardless of $(u,v)\in\mathbb{R}^n\times\mathbb{R}^n$.

On the other hand, by applying (1. 2) with $\mathbf{a} = n[\frac{\alpha+\beta}{n+1}]$ and $\mathbf{N} = n$, we find

$$\left\{\int_{\mathbb{R}^n}\left\{\int_{\mathbb{R}^n}|u-\xi|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}\left\|f(\xi,\eta,\cdot)\right\|_{\mathbf{L}^p(\mathbb{R})}d\xi\right\}^q du\right\}^{\frac{1}{q}} \le \mathfrak{B}_{p\,q}\left\{\int_{\mathbb{R}^n}\left\|f(u,\eta,\cdot)\right\|^p_{\mathbf{L}^p(\mathbb{R})}du\right\}^{\frac{1}{p}} \qquad (3.\ 8)$$

and

$$\left\{\int_{\mathbb{R}^n}\left\{\int_{\mathbb{R}^n}|v-\eta|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}\left\|f(\xi,\eta,\cdot)\right\|_{\mathbf{L}^p(\mathbb{R})}d\eta\right\}^q dv\right\}^{\frac{1}{q}} \le \mathfrak{B}_{p\,q}\left\{\int_{\mathbb{R}^n}\left\|f(\xi,v,\cdot)\right\|^p_{\mathbf{L}^p(\mathbb{R})}dv\right\}^{\frac{1}{p}}. \qquad (3.\ 9)$$

From (3. 6), we have

$$\begin{aligned}
&\left\|\mathbf{I}_{\alpha\beta\vartheta}f\right\|_{\mathbf{L}^q(\mathbb{R}^{2n+1})} \\
&\le \left\{\iiint_{\mathbb{R}^{2n+1}}\left\{\iint_{\mathbb{R}^{2n}}|u-\xi|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}|v-\eta|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}\mathbf{F}_{\alpha\beta}(\xi,\eta,u,v,t)d\xi d\eta\right\}^q dudvdt\right\}^{\frac{1}{q}} \\
&\le \left\{\iint_{\mathbb{R}^{2n}}\left\{\iint_{\mathbb{R}^{2n}}|u-\xi|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}|v-\eta|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}\left\{\int_{\mathbb{R}}\mathbf{F}^q_{\alpha\beta}(\xi,\eta,u,v,t)dt\right\}^{\frac{1}{q}}d\xi d\eta\right\}^q dudv\right\}^{\frac{1}{q}} \\
&\qquad\qquad \text{by Minkowski integral inequality} \\
&\le \mathfrak{B}_{p\,q}\left\{\iint_{\mathbb{R}^{2n}}\left\{\iint_{\mathbb{R}^{2n}}|u-\xi|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}|v-\eta|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}\left\|f(\xi,\eta,\cdot)\right\|_{\mathbf{L}^p(\mathbb{R})}d\xi d\eta\right\}^q dudv\right\}^{\frac{1}{q}} \quad \text{by (3. 7)} \\
&\le \mathfrak{B}_{p\,q}\left\{\int_{\mathbb{R}^n}\left\{\int_{\mathbb{R}^n}\left\{\int_{\mathbb{R}^n}|v-\eta|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}\left\|f(u,\eta,\cdot)\right\|_{\mathbf{L}^p(\mathbb{R})}d\eta\right\}^p du\right\}^{\frac{q}{p}}dv\right\}^{\frac{1}{q}} \quad \text{by (3. 8)} \\
&\le \mathfrak{B}_{p\,q}\left\{\int_{\mathbb{R}^n}\left\{\int_{\mathbb{R}^n}\left\{\int_{\mathbb{R}^n}|v-\eta|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}\left\|f(u,\eta,\cdot)\right\|_{\mathbf{L}^p(\mathbb{R})}d\eta\right\}^q dv\right\}^{\frac{p}{q}}du\right\}^{\frac{1}{p}} \\
&\qquad\qquad \text{by Minkowski integral inequality} \\
&\le \mathfrak{B}_{p\,q}\left\{\iint_{\mathbb{R}^{2n}}\left\|f(u,v,\cdot)\right\|^p_{\mathbf{L}^p(\mathbb{R})}dudv\right\}^{\frac{1}{p}} \quad \text{by (3. 9)} \\
&= \mathfrak{B}_{p\,q}\left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})}.
\end{aligned} \qquad (3.\ 10)$$

4 Proof of Theorem Three

Recall $\mathbf{M}_\gamma$ defined in (1. 14) for $0 \le \gamma < 1$. By taking $\xi \longrightarrow u-\xi$, $\eta \longrightarrow v-\eta$ and $\tau \longrightarrow t-\tau$, $\mathbf{M}_\gamma$ can be equivalently defined as

$$\mathbf{M}_\gamma f(u,v,t) = \sup_{\mathbf{R}\ni(u,v,t)} \mathbf{vol}\{\mathbf{R}\}^{\gamma-1} \iiint_{\mathbf{R}} \Big| f(\xi,\eta,\tau+\mu(u\cdot\eta - v\cdot\xi))\Big| d\xi d\eta d\tau. \qquad (4.\ 11)$$

Similarly, $\mathbf{M}_{\alpha\beta}$ defined in (1. 11) is equivalent to

$$\begin{aligned}\mathbf{M}_{\alpha\beta} f(u,v,t) &= \sup_{\mathbf{R}\ni(u,v,t):\mathbf{vol}\{I\}=\mathbf{vol}\{\mathbf{Q}\}^{\frac{1}{n}}\mathbf{vol}\{\mathbf{P}\}^{\frac{1}{n}}} \\ &\mathbf{vol}\{\mathbf{Q}\}^{\frac{\alpha}{n}-1}\mathbf{vol}\{\mathbf{P}\}^{\frac{\alpha}{n}-1}\mathbf{vol}\{I\}^{\beta-1} \iiint_{\mathbf{R}} \Big| f(\xi,\eta,\tau+\mu(u\cdot\eta - v\cdot\xi))\Big| d\xi d\eta d\tau\end{aligned} \qquad (4.\ 12)$$

where $\mathbf{R} = \mathbf{Q}\times\mathbf{P}\times\mathbf{I} \subset \mathbb{R}^n\times\mathbb{R}^n\times\mathbb{R}$.

Note that $\mathbf{vol}\{I\} = \mathbf{vol}\{\mathbf{Q}\}^{\frac{1}{n}}\mathbf{vol}\{\mathbf{P}\}^{\frac{1}{n}}$ implies $\mathbf{vol}\{\mathbf{R}\} = \mathbf{vol}\{\mathbf{Q}\}^{1+\frac{1}{n}}\mathbf{vol}\{\mathbf{P}\}^{1+\frac{1}{n}}$. From (4. 12), we find

$$\begin{aligned}\mathbf{M}_{\alpha\beta} f(u,v,t) &= \sup_{\mathbf{R}\ni(u,v,t):\mathbf{vol}\{I\}=\mathbf{vol}\{\mathbf{Q}\}^{\frac{1}{n}}\mathbf{vol}\{\mathbf{P}\}^{\frac{1}{n}}} \\ &\mathbf{vol}\{\mathbf{Q}\}^{\left[\frac{\alpha+\beta}{n+1}-1\right](1+\frac{1}{n})}\mathbf{vol}\{\mathbf{P}\}^{\left[\frac{\alpha+\beta}{n+1}-1\right](1+\frac{1}{n})} \iiint_{\mathbf{R}} \Big| f(\xi,\eta,\tau+\mu(u\cdot\eta - v\cdot\xi))\Big| d\xi d\eta d\tau \\ &= \sup_{\mathbf{R}\ni(u,v,t):\mathbf{vol}\{I\}=\mathbf{vol}\{\mathbf{Q}\}^{\frac{1}{n}}\mathbf{vol}\{\mathbf{P}\}^{\frac{1}{n}}} \mathbf{vol}\{\mathbf{R}\}^{\frac{\alpha+\beta}{n+1}-1} \iiint_{\mathbf{R}} \Big| f(\xi,\eta,\tau+\mu(u\cdot\eta - v\cdot\xi))\Big| d\xi d\eta d\tau \qquad (4.\ 13) \\ &\le \sup_{\mathbf{R}\ni(u,v,t)} \mathbf{vol}\{\mathbf{R}\}^{\gamma-1} \iiint_{\mathbf{R}} \Big| f(\xi,\eta,\tau+\mu(u\cdot\eta - v\cdot\xi))\Big| d\xi d\eta d\tau \qquad \left(\gamma = \tfrac{\alpha+\beta}{n+1}\right) \\ &= \mathbf{M}_\gamma f(u,v,t).\end{aligned}$$

Hence, $\mathbf{M}_{\alpha\beta}$ is controlled by the strong fractional maximal operator $\mathbf{M}_\gamma$ whenever $\gamma = \frac{\alpha+\beta}{n+1}$. Let $\gamma = \frac{1}{p}-\frac{1}{q}$, $1 < p \le q < \infty$. In order to prove the converse, we need the following multi-parameter covering lemma.

Córdoba-Fefferman covering lemma

Let $\left\{\mathbf{R}_j\right\}_{j=1}$ *be a collection of rectangles in* $\mathbb{R}^{2n+1}$ *parallel to the coordinates. There is a subsequence* $\left\{\widehat{\mathbf{R}}_k\right\}_{k=1}$ *such that*

$$\mathbf{vol}\left\{\bigcup_j \mathbf{R}_j\right\} \lesssim \mathbf{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\} \qquad (4.\ 14)$$

and

$$\left\|\sum_k \chi_{\widehat{\mathbf{R}}_k}\right\|^p_{\mathbf{L}^p(\mathbb{R}^{2n+1})} \le \mathfrak{B}_p\, \mathbf{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\}, \qquad 1<p<\infty \qquad (4.\ 15)$$

where χ *is an indicator function.*

Remark 4.1. *This covering lemma is established by Córdoba and Fefferman [3] within a much more general setting. Namely, the Lebesgue measure is replaced by an absolutely continuous measure satisfying the rectangle A_∞-property. In the following subsection, we prove this covering lemma w.r.t Lebesgue measure more directly.*

Define

$$\mathbf{U}_\lambda = \Big\{(u,v,t)\in\mathbb{R}^n\times\mathbb{R}^n\times\mathbb{R}\colon \mathbf{M}_\gamma f(u,v,t)>\lambda\Big\}. \tag{4. 16}$$

Given any $(u,v,t)\in\mathbf{U}_\lambda$, there is a rectangle $\mathbf{R}_j\ni(u,v,t)$ such that

$$\mathbf{vol}\{\mathbf{R}_j\}^{\gamma-1}\iiint_{\mathbf{R}_j}\Big|f(\xi,\eta,\tau+\mu(u\cdot\eta-v\cdot\xi))\Big|d\xi d\eta d\tau > \frac{1}{2}\lambda. \tag{4. 17}$$

Let (u,v,t) run through the set $\mathbf{U}_\lambda$. We have

$$\mathbf{U}_\lambda \subset \bigcup_j \mathbf{R}_j. \tag{4. 18}$$

By applying the covering lemma, we select a subsequence $\{\widehat{\mathbf{R}}_k\}_{k=1}^\infty$ from the union above and

$$\mathbf{vol}\Big\{\mathbf{U}_\lambda\Big\} \lesssim \mathbf{vol}\Big\{\bigcup_j\mathbf{R}_j\Big\} \lesssim \mathbf{vol}\Big\{\bigcup_k\widehat{\mathbf{R}}_k\Big\} \qquad \text{by (4. 14)}. \tag{4. 19}$$

Furthermore, we have

$$\begin{aligned}
\mathbf{vol}\Big\{\bigcup_k\widehat{\mathbf{R}}_k\Big\} &\le \sum_k \mathbf{vol}\Big\{\widehat{\mathbf{R}}_k\Big\}\\
&\lesssim \lambda^{-\frac{1}{1-\gamma}}\sum_k\left\{\iiint_{\widehat{\mathbf{R}}_k}\Big|f(\xi,\eta,\tau+\mu(u\cdot\eta-v\cdot\xi))\Big|d\xi d\eta d\tau\right\}^{\frac{1}{1-\gamma}} \qquad \text{by (4. 17)}\\
&\lesssim \lambda^{-\frac{1}{1-\gamma}}\left\{\sum_k\iiint_{\widehat{\mathbf{R}}_k}\Big|f(\xi,\eta,\tau+\mu(u\cdot\eta-v\cdot\xi))\Big|d\xi d\eta d\tau\right\}^{\frac{1}{1-\gamma}} \qquad (0\le\gamma<1)\\
&= \lambda^{-\frac{1}{1-\gamma}}\left\{\iiint_{\mathbb{R}^{2n+1}}\left|f(\xi,\eta,\tau+\mu(u\cdot\eta-v\cdot\xi))\sum_k\chi_{\widehat{\mathbf{R}}_k}(\xi,\eta,\tau)\right|d\xi d\eta d\tau\right\}^{\frac{1}{1-\gamma}} \qquad (4.\ 20)\\
&\le \lambda^{-\frac{1}{1-\gamma}}\left\{\iiint_{\mathbb{R}^{2n+1}}\Big|f(\xi,\eta,\tau+\mu(u\cdot\eta-v\cdot\xi))\Big|^p d\xi d\eta d\tau\right\}^{\frac{1}{p}\frac{1}{1-\gamma}}\left\|\sum_k\chi_{\widehat{\mathbf{R}}_k}\right\|^{\frac{1}{1-\gamma}}_{\mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^{2n+1})}\\
&\qquad\text{by Hölder inequality}\\
&= \lambda^{-\frac{1}{1-\gamma}}\left\{\iint_{\mathbb{R}^{2n}}\big\|f(\xi,\eta,\cdot)\big\|^p_{\mathbf{L}^p(\mathbb{R})}d\xi d\eta\right\}^{\frac{1}{p}\frac{1}{1-\gamma}}\left\|\sum_k\chi_{\widehat{\mathbf{R}}_k}\right\|^{\frac{1}{1-\gamma}}_{\mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^{2n+1})}\\
&\le \lambda^{-\frac{1}{1-\gamma}}\big\|f\big\|^{\frac{1}{1-\gamma}}_{\mathbf{L}^p(\mathbb{R}^{2n+1})}\mathbf{vol}\Big\{\bigcup_k\widehat{\mathbf{R}}_k\Big\}^{\frac{p-1}{p}\frac{1}{1-\gamma}} \qquad \text{by (4. 15)}.
\end{aligned}$$

By raising both sides of (4. 20) to the $(1-\gamma)$-th power and then taking into account for $1-\gamma-\frac{p-1}{p}=\frac{1}{p}-\left[\frac{1}{p}-\frac{1}{q}\right]=\frac{1}{q}$, we find

$$\mathbf{vol}\Big\{\bigcup_k \widehat{\mathbf{R}}_k\Big\}^{\frac{1}{q}} \lesssim \frac{1}{\lambda}\left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})}. \tag{4. 21}$$

Let $\mathbf{U}_\lambda$ defined in (4. 16). From (4. 19) and (4. 21), we obtain

$$\begin{aligned}\mathbf{vol}\Big\{(u,v,t)\in\mathbb{R}^n\times\mathbb{R}^n\times\mathbb{R}\colon \mathbf{M}_\gamma f(u,v,t)>\lambda\Big\}^{\frac{1}{q}} &\lesssim \mathbf{vol}\Big\{\bigcup_k \widehat{\mathbf{R}}_k\Big\}^{\frac{1}{q}}\\ &\lesssim \frac{1}{\lambda}\left\|f\right\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})}.\end{aligned} \tag{4. 22}$$

By using this weak type (p,q)-estimate and applying Marcinkiewicz interpolation theorem, we conclude that $\mathbf{M}_\gamma$ is bounded from $\mathbf{L}^p(\mathbb{R}^{2n+1})$ to $\mathbf{L}^q(\mathbb{R}^{2n+1})$ if $\gamma=\frac{1}{p}-\frac{1}{q}, 1<p\le q<\infty$.

4.1 A simpler proof of the covering lemma w.r.t Lebesgue measure

We select $\widehat{\mathbf{R}}_k$ from $\left\{\mathbf{R}_j\right\}_{j=1}$ as follows. Let $\widehat{\mathbf{R}}_1=\mathbf{R}_1$. Having chosen $\widehat{\mathbf{R}}_1,\widehat{\mathbf{R}}_2,\ldots,\widehat{\mathbf{R}}_{k-1}$, we pick $\widehat{\mathbf{R}}_k$ as the first rectangle $\mathbf{R}$ on the list of $\mathbf{R}_j$'s after $\widehat{\mathbf{R}}_{k-1}$ so that

$$\mathbf{vol}\left\{\mathbf{R}\cap\bigcup_{\ell=1}^{k-1}\widehat{\mathbf{R}}_\ell\right\} \le \frac{1}{2}\mathbf{vol}\left\{\mathbf{R}\right\}. \tag{4. 23}$$

Suppose that $\mathbf{R}$ is an unselected rectangle. There is a positive number N such that $\mathbf{R}$ is on the list of $\mathbf{R}_j$'s after $\widehat{\mathbf{R}}_N$ and

$$\mathbf{vol}\left\{\mathbf{R}\cap\bigcup_{k=1}^{N}\widehat{\mathbf{R}}_k\right\} > \frac{1}{2}\mathbf{vol}\left\{\mathbf{R}\right\}. \tag{4. 24}$$

Let $\mathbf{M}$ be the strong maximal operator defined in $\mathbb{R}^{2n+1}$ which is $\mathbf{L}^p$-bounded for $1<p<\infty$. From (4. 24), we find

$$\mathbf{M}\chi_{\bigcup_k\widehat{\mathbf{R}}_k}(u,v,t) > \frac{1}{2}, \qquad (u,v,t)\in\bigcup_j\mathbf{R}_j. \tag{4. 25}$$

By applying the $\mathbf{L}^2$-boundedness of $\mathbf{M}$, we have

$$\begin{aligned}\mathbf{vol}\Big\{\bigcup_j\mathbf{R}_j\Big\} &= \iiint_{\bigcup_j\mathbf{R}_j} dudvdt\\ &\le 2^2\iiint_{\bigcup_j\mathbf{R}_j}\left[\mathbf{M}\chi_{\bigcup_k\widehat{\mathbf{R}}_k}\right]^2(u,v,t)dudvdt \qquad \text{by (4. 25)}\\ &\lesssim \iiint_{\mathbb{R}^{2n+1}}\left[\chi_{\bigcup_k\widehat{\mathbf{R}}_k}\right]^2(u,v,t)dudvdt\\ &= \iiint_{\bigcup_k\widehat{\mathbf{R}}_k} dudvdt = \mathbf{vol}\Big\{\bigcup_k\widehat{\mathbf{R}}_k\Big\}.\end{aligned} \tag{4. 26}$$

On the other hand, (4. 23) suggests

$$\mathbf{vol}\left\{\widehat{\mathbf{R}}_k\cap\bigcup_{\ell=1}^{k-1}\widehat{\mathbf{R}}_\ell\right\} ~\leq~ \frac{1}{2}\mathbf{vol}\left\{\widehat{\mathbf{R}}_k\right\},\qquad k=1,2,\ldots \tag{4. 27}$$

which further implies

$$\mathbf{vol}\left\{\widehat{\mathbf{R}}_k\setminus\bigcup_{\ell=1}^{k-1}\widehat{\mathbf{R}}_\ell\right\} ~>~ \frac{1}{2}\mathbf{vol}\left\{\widehat{\mathbf{R}}_k\right\},\qquad k=1,2,\ldots. \tag{4. 28}$$

Let $\phi\in\mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^{2n+1})$ and $\left\|\phi\right\|_{\mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^{2n+1})}=1$. We have

$$\begin{aligned}
&\iiint_{\mathbb{R}^{2n+1}}\phi(\xi,\eta,\tau)\sum_k\chi_{\widehat{\mathbf{R}}_k}(\xi,\eta,\tau)d\xi d\eta d\tau\\
&=~\sum_k\iiint_{\widehat{\mathbf{R}}_k}\phi(\xi,\eta,\tau)d\xi d\eta d\tau\\
&=~\sum_k\left\{\mathbf{vol}\{\widehat{\mathbf{R}}_k\}^{-1}\iiint_{\widehat{\mathbf{R}}_k}\phi(\xi,\eta,\tau)d\xi d\eta d\tau\right\}\mathbf{vol}\left\{\widehat{\mathbf{R}}_k\right\}\\
&<~2\sum_k\left\{\mathbf{vol}\{\widehat{\mathbf{R}}_k\}^{-1}\iiint_{\widehat{\mathbf{R}}_k}|\phi(\xi,\eta,\tau)|d\xi d\eta d\tau\right\}\mathbf{vol}\left\{\widehat{\mathbf{R}}_k\setminus\bigcup_{\ell=1}^{k-1}\widehat{\mathbf{R}}_\ell\right\}\qquad\text{by (4. 28)}\qquad(4.\ 29)\\
&\lesssim~\sum_k\int_{\widehat{\mathbf{R}}_k\setminus\bigcup_{\ell=1}^{k-1}\widehat{\mathbf{R}}_\ell}\left\{\mathbf{vol}\{\widehat{\mathbf{R}}_k\}^{-1}\iiint_{\widehat{\mathbf{R}}_k}|\phi(\xi,\eta,\tau)|d\xi d\eta d\tau\right\}dudvdt\\
&\leq~\sum_k\iiint_{\widehat{\mathbf{R}}_k\setminus\bigcup_{\ell=1}^{k-1}\widehat{\mathbf{R}}_\ell}\mathbf{M}\phi(u,v,t)dudvdt\\
&=~\iiint_{\bigcup_k\widehat{\mathbf{R}}_k}\mathbf{M}\phi(u,v,t)dudvdt.
\end{aligned}$$

Hölder inequality further implies

$$\begin{aligned}
\iiint_{\bigcup_k\widehat{\mathbf{R}}_k}\mathbf{M}\phi(u,v,t)dudvdt ~&\leq~ \left\|\mathbf{M}\phi\right\|_{\mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^{2n+1})}\mathbf{vol}\left\{\bigcup_k\widehat{\mathbf{R}}_k\right\}^{\frac{1}{p}}\\
&\leq~\mathfrak{B}_p\left\|\phi\right\|_{\mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^{2n+1})}\mathbf{vol}\left\{\bigcup_k\widehat{\mathbf{R}}_k\right\}^{\frac{1}{p}}\qquad\mathbf{M}\text{ bounded on }\mathbf{L}^p(\mathbb{R}^{2n+1})\\
&=~\mathfrak{B}_p\,\mathbf{vol}\left\{\bigcup_k\widehat{\mathbf{R}}_k\right\}^{\frac{1}{p}}.
\end{aligned} \tag{4. 30}$$

By substituting (4. 30) to (4. 29) and taking the supremum of ϕ, we arrive at

$$\left\|\sum_k\chi_{\widehat{\mathbf{R}}_k}\right\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})} ~\leq~ \mathfrak{B}_p\,\mathbf{vol}\left\{\bigcup_k\widehat{\mathbf{R}}_k\right\}^{\frac{1}{p}},\qquad 1<p<\infty. \tag{4. 31}$$

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